

ELECTRONIC, ELECTRICAL & SYSTEMS ENGINEERING

**UNIVERSITY OF
BIRMINGHAM**

BEng Final Year Project (EE3P)

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**Project Title: Design and Build of a Dual-Core Music
Synthesizer for Educational Enhancement**

Supervisor: Dr. Timothy Jackson

Project Self Assessment

Checklist: Put a “Y” in the column which corresponds to your assessment of our own ability

Category	I find this very difficult	I find this a bit difficult	neutral	I find this fairly easy	I find this very easy
Ability to work independently				Y	
Ability to manage my time					Y
Ability to learn new skills or concepts in depth				Y	
Ability to learn new concepts or skills quickly			Y		
Ability to focus on targets					Y
Ability to apply things that I have learned				Y	
Ability to understand the implications of results and findings		Y			
Ability to draw conclusions			Y		

What aspects of your project did you enjoy and or went well (up to 50 words)	I found the iterative process of building each part of the design particularly rewarding. I was able to recall and apply concepts I had learned as well as new knowledge I acquired during the process. I also enjoyed working independently due to clear goals outlined during meetings with my supervisor.
What aspects of your project did you find difficult or would you change (up to 50 words)	I found difficulty with relating the eventual results with the theme of the project considering the lack of abundant academic material on this topic. If I were to work on this project again, I would outline specific educational outcomes before the design phase of the hardware.

Acknowledgements

I would like to express my gratitude towards my project supervisor, Dr. Timothy Jackson for his thorough interest in guiding me during this project. His desire to use his extensive knowledge and experience for my support as well as the provision of actionable feedback, helped me to produce a system of which I am proud.

Abstract

Analogue electrical concepts often prove difficult to understand for engineering students. This cognitive hurdle requires an interactive educational medium to bridge the theoretical to practical gap of understanding analogue electronics components and systems. The primary aim of this project is to produce a system that helps to overcome this hurdle. This project entails the design, construction, and evaluation of a bespoke dual-core music synthesizer to function as that interactive medium. A music synthesizer is perfect for this application because it clearly encapsulates how separate electronic subsystems can operate together as part of a complete system. The project modelled the system after a subtractive synthesis architecture with the inclusion of fundamental analogue components forming the subsystems. These subsystems include oscillation, mixing, filtering, and output attenuation stages.

Physical implementation produced a fully functional prototype with the inclusion of the separate subsystems. Oscilloscope testing analysed the performance of each individual stage. Subsequent data extraction and plotting produced graphs for comparative analysis against theoretical materials to verify the expected behaviour of each subsystem.

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List of Acronyms

VCO: Voltage-Controlled Oscillator

VCF: Voltage-Controlled Filter

VCA: Voltage-Controlled Amplifier

Op-amp: Operational Amplifier

I/O: Input and Output

J-FET: Joint Field-Effect Transistor

MOSFET: Metal-Oxide-Semiconductor Field-Effect Transistor

SPI: Serial Peripheral Interface

IDE: Integrated Development Environment

FFT: Fast Fourier Transform

LED: Light Emitting Diode

Digipot: Digital Potentiometer

Chapter 1: Introduction

1.1 Project Motivation

A notable percentage of engineering students struggle to relate the theoretical concepts they learn to a more practical environment (Billis, et al., 2011). A large part of this educational difficulty lies in the lack of a practical system-level medium to properly demonstrate how physical implementations of these theoretical concepts operate in cohesion. A music synthesizer provides a suitable solution to improve understanding of electrical constructs through interactive demonstration, hence the motivation behind this project.

An analogue music synthesizer is appropriate for this purpose due to its general functional block architecture with each subsystem contributing a specified purpose to the general system. These include VCOs, VCFs, and VCAs operating in cohesion. By observing oscilloscope readings and resulting audio from this system, students can verify the practical implications of theoretical analogue electronics concepts.

1.2 Project Aims and Objectives

The primary aim of the project is to design and build a dual-core music synthesizer as an educational tool to improve understanding in the study of analogue electronic concepts. The following objectives are to:

- Define the music synthesizer functional block architecture.
- Design and simulate the fundamental analogue subsystems.
- Develop a digital control interface for the system.
- Construct a fully operational physical prototype based on the design.
- Evaluate the system functionality through oscilloscope waveform analysis.

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1.3 Literature Review

This subheading reviews relevant academic literature and engineering principles supporting the design and development of the music synthesizer for educational purposes. The review evaluates the effectiveness of the synthesizer as an educational tool, the fundamental hardware architecture and subsystems required, and methodologies for integrating digital microcontrollers to achieve accurate control over the system. The theory explored here directly informs hardware and software design choices detailed further in this report.

1.3.1 The Music Synthesizer as an Educational Instrument

Pagiatakis and Voudoukis (2017) pointed out that students often find difficulty in understanding foundational concepts of electronics including the principles of op-amp structures, stating: ‘...it has been observed that the students tend to address electronic circuit problems in a rather local and sequential way with no reference to the appropriate principles, rules, and methods.’. This research suggests that students require an active medium allowing them to gain understanding through quantitative observations.

Billis et al. (2011) provide a viable suggestion for this active medium. They highlight that music synthesizers comprehensively encapsulate intricate electronic concepts into a single functional system. By varying the parameters of the fundamental op-amp circuits used to construct the synthesizer, students can observe for themselves the practical applications of their theoretical study. Therefore, this project adopts a music synthesizer model to provide this active learning medium.

1.3.2 Synthesizer Architecture

A logical framework of functional blocks effectively demonstrates the stages of analogue signal manipulation. Malakçı and Erdoğan (2021) outline a subtractive synthesis architecture with its fundamental building blocks. This structure typically consists of VCOs, VCFs, and VCAs.

Introduction

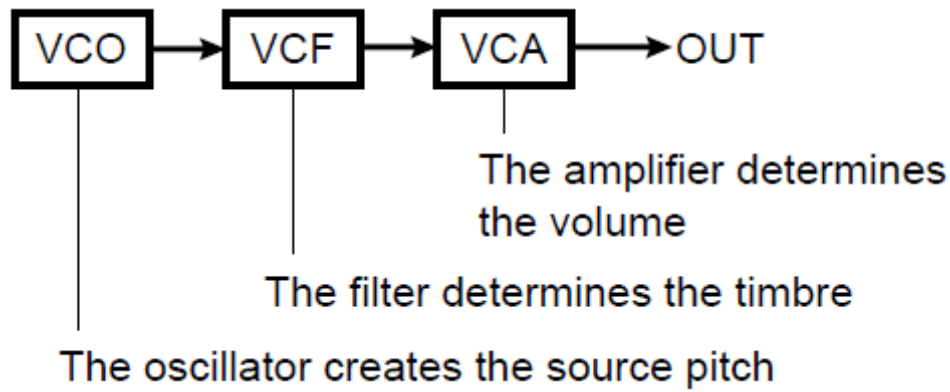


Figure 1. Fundamental blocks of a music synthesizer (Mister, B., 2018)

The process flow begins with the generation of harmonically rich waveforms by the VCO. Roma (2024) establishes that distinct generated waveforms can combine into one signal through mixers modelled as summing amplifiers to produce an enriched audio signal.

Subsequent to generation and mixing, the signal is 'shaped' by a filter. A synthesizer circuit derives its primary sonic identity from the filtering stage. The filter removes specified harmonic frequencies from the signal, hence altering its timbre. Kladovščikov et al. (2019) outlined the advantages of selecting the Sallen-Key architecture for the filter topology.

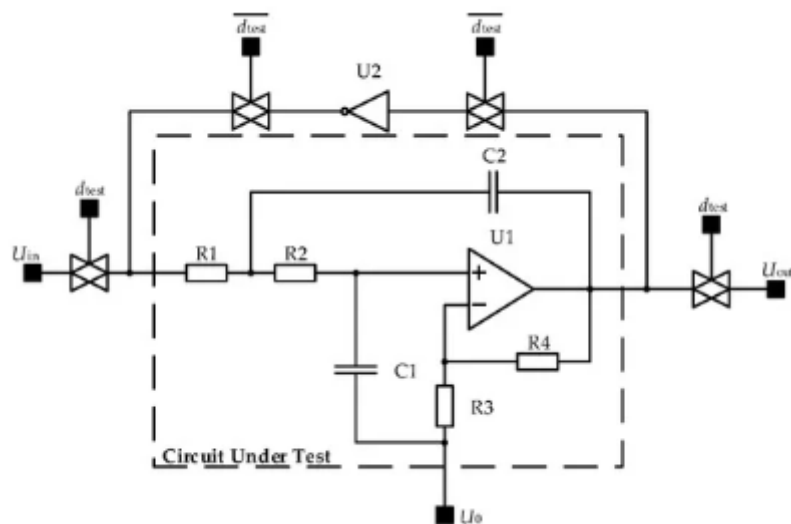


Figure 2. Structure of the second-order Sallen-Key low pass filter (Kladovščikov et al., 2019)

The governing transfer function of the second order Sallen-Key low pass filter is as follows:

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$$H_{SK}(s) = \frac{\left(1 + \frac{R_4}{R_3}\right) \frac{1}{C_1 C_2 R_1 R_2}}{s^2 + s \left(\frac{1}{C_1 R_1} + \frac{1}{C_1 R_2} + \frac{R_4}{C_2 R_2 R_3} \right) + \frac{1}{C_1 C_2 R_1 R_2}} \quad (1.1)$$

The literature demonstrates the need for less passive and active components for this specific architecture as compared to other designs such as the Tow-Thomas or Akerberg-Mossberg filters. In addition, Kladovščikov et al. (2019) show quantitatively that a Sallen-Key configuration demonstrates characteristics of extremely narrow and undetectable fault margins as defined by Parametric Test Limitations proving good operational reliability, hence presenting advantages for use in the synthesizer prototype.

1.3.3 Microcontroller Integration

Music synthesizer designs usually implement digital components to increase the operational ability of the analogue circuits. Choqueuse et al. (2018) demonstrated effective integration of microcontrollers to construct digital synthesizers used in educational applications. This literature verifies the precise sequencing capabilities provided by digital control layers over analogue signal generation stages. This study detailed tasks given to students of developing C programs to drive an STM32 microcontroller to handle Musical Instrument Digital Interface parsing, as well as other synthesizer related control functions. The results demonstrated improved understanding of digital processing operations among participant students. This project adapts the programming concepts outlined by Choqueuse (2018) while maintaining a more physical application identified by Billis et al (2011). Expanding on this control methodology, Kvalsund and Winkler (2023) highlight that Arduino microcontrollers deliver an efficient solution capable of replacing expensive control systems through its distinct features which are high flexibility, and low hardware cost. The research further demonstrates the Arduino's effectiveness in bridging digital programs with physical hardware systems through the microcontroller's I/O terminals. By utilising an Arduino to drive the circuit, the system can bridge a C programming logic with physical components, offering students an interactive control interface.

Chapter 2: System Design Methodology

This chapter focuses on design choices of the music synthesizer and their respective justifications. The design adopts a strict methodology which ensures cohesive operation of the separate analogue subsystems with digital control inputs. The design choices underwent simulations using the NI Multisim software to visualize circuit behaviour before physical implementation. Industry standard simulation and vast component library set this software above others.

2.1 System Overview

The analogue section of the system makes modifications around the fundamentals of the subtractive synthesis architecture outlined in the previous chapter (Malakçı and Erdoğan, 2021). Two independent VCOs form the signal generation stage, which concurrently produce distinct fundamental pulse and triangular waveforms. This decision allows the system to effectively demonstrate the mixing capabilities of the summing amplifiers, which combine the triangular waveforms with the pulse waveforms from both oscillators to create a single, harmonically dense, audio signal. Following the mixing stage, the signal passes through an active filter which aggressively ‘smoothens’ the signal by removing its higher frequency harmonic contents, hence, defining the final audio timbre. Finally, the signal reaches the amplification stage for volume control.

The digital section consists of an Arduino microcontroller which provide precise sequential commands. These commands control a digital potentiometer which regulates the DC voltages directed to the signal generation stage.

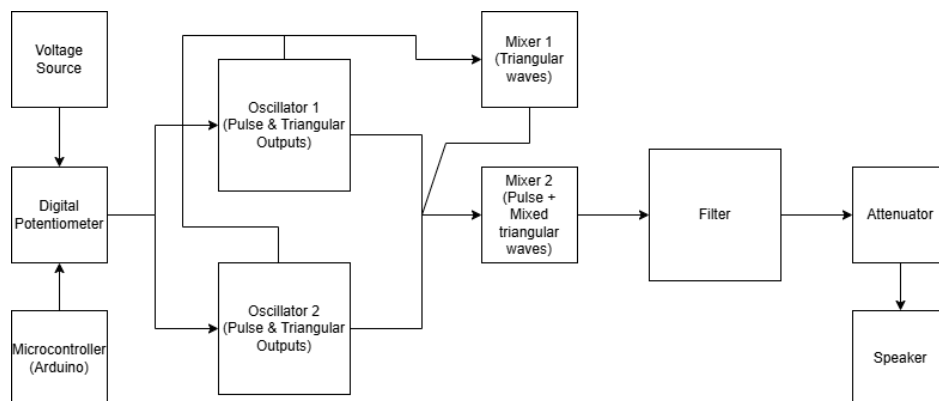


Figure 3. Block Diagram of System Architecture

System Design Methodology

2.1.1 Educational Influence on Design

The overall system design meticulously considered the educational value encapsulated by its architecture. The design incorporates important analogue electronic concepts as outlined in the literature from the previous chapter, taught over the course of a BEng Electronic and Electrical Engineering degree.

Table 1. Applied Educational Concepts to Synthesizer Design

Educational Concept	Synthesizer Stage	BEng Degree Module
Oscillation using op-amps	Signal Generation	Analogue Electronics (2 nd Year)
Combining Analogue Signals using op-amps	Signal Combination	
Active filters & Transfer Functions	Filtering	Analogue Electronics (3 rd Year)
Op-amp amplification & Inversion	Attenuation	Analogue Electronics (1 st & 2 nd Year)
Microcontroller Control & Implementation	Digital Control	Digital Electronics (3 rd Year)

The above table shows the principles of the music synthesizer design at each functional stage and how its educational applications span over the course of the BEng degree.

2.2 Op-amp Selection

The creation of analogue circuits related to audio processing heavily relies on op-amps due to their versatile functionality and superior performance in comparison to passive circuits. They enable precise signal conditioning through the minimisation of resonance distortions (Carter and Brown, 2001). Hence, the need for op-amps in each functional stage is imperative.

The TL082CN op-amp datasheet details a J-FET input model providing a high input impedance of 1 T Ω , preventing the analogue components from drawing excess current and reducing risk of damage. The physical chip also contains two separate op-amps which minimises the required number of components to construct the system.

The TL082CN op-amp also proved to be a suitable option based on the system's requirements. The op-amp's slew rate of 13 volts per microsecond as defined in its datasheet, allows for rapid voltage transitions to accurately generate high-frequency pulse and triangular waves.

Table 2. Comparison of TL082CN and LM741 Op-Amps

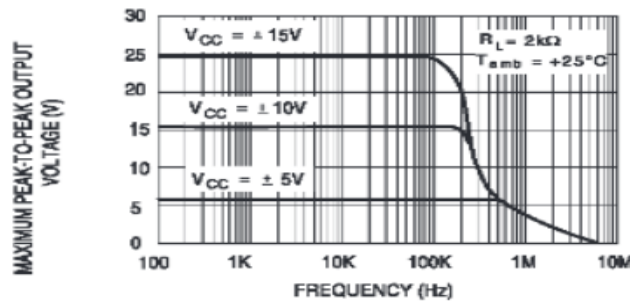
Parameter	TL082CN	LM741
Slew Rate (V/ μ s)	13	0.5
Input Impedance (Ω)	1 T	2 M

The table above shows the superiority of the TL082CN design compared to the LM741 used various school laboratory sessions.

The formula below determines the effect of an op-amp's slew rate on its maximum operating frequency.

$$f_{max} = \frac{Slew\ Rate}{2\pi.V_{peak}} \quad (2.1)$$

Assuming a peak voltage of 15 V, the maximum operating frequency of the TL082CN would be 137 931 Hz compared to the LM741's maximum of just 5305 Hz. This calculation proves the TL082CN is far more capable of tracking rapid voltage changes than the LM741.



**Figure 4. Max. Peak-to-Peak Output Voltage v Frequency of TL082CN
(STMicroelectronics, 2001)**

The figure above shows the TL082CN can attain high output frequencies at different voltage levels which is advantageous for the audio range of the synthesizer.

2.3 Signal Generation

Xu, et al. (2021) documented the concept of using a combined integrator and Schmitt trigger architecture for analogue signal generation. The literature details the advantages of this architecture concerning constant power consumption and waveform integrity. The architecture also entails a minimal operating linearity error of 1.5% allowing a higher degree of control accuracy.

Two identical versions of the defined VCOs employed an integrator circuit to drive a Schmitt trigger to produce continuous and stable oscillations.

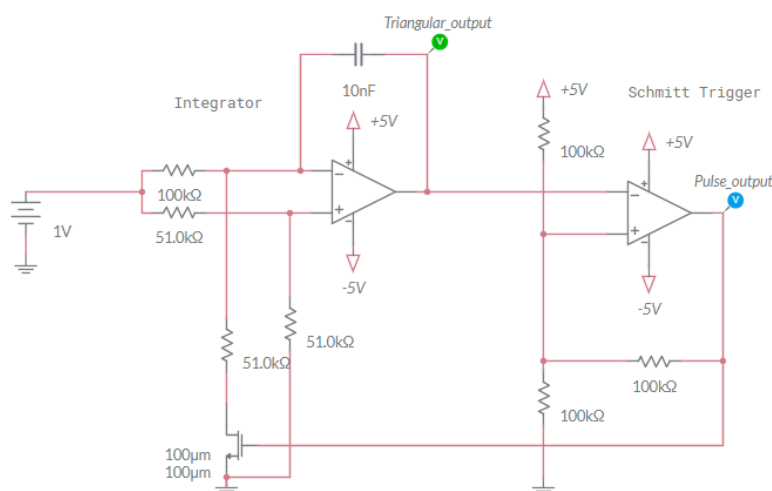


Figure 5. Voltage-Controlled Oscillator Schematic

2.3.1 Integrator Operation

The integrator converts the DC voltage input into a linearly charging output or ‘ramp.’

The frequency of the resulting waveform is inversely proportional to the capacitance in the feedback loop ($f \propto 1/C$). To increase the baseline frequency of the synthesizer’s signal generation to a more useful musical range, a 2.2 nF capacitor replaces the conventional 10 nF capacitor through empirical means. Theoretically, reducing the capacitance by a factor of 4.54 would increase the oscillation frequency by the same factor. This capacitance set a viable frequency setting of approximately 100 Hz per one volt input sitting nicely within the human auditory range. The 100 k Ω resistor regulates the charging current at the input.

2.3.2 Schmitt Trigger Operation

The Schmitt trigger receives the integrator output to generate the pulse waveform. Due to hysteresis, the Schmitt trigger creates two threshold values to compare to the input signal. An upper and lower threshold value based on the reference voltage provided by a potential divider made with identical 100 k Ω resistors. The two threshold values resulted from feeding the output back into the op-amp’s positive input through a 100 k Ω resistor. The Schmitt trigger constantly monitors the triangular output from the integrator and snaps the op-amp output high (+5 V) when the ‘ramp’ hits the upper threshold value.

2.3.3 Completion of the Loop

The output of the Schmitt trigger connected back to the system to form a loop through an n-channel MOSFET. The high output from the Schmitt trigger causes the MOSFET to turn on. This switching action causes the vertical 51 k Ω resistors to connect to ground. Hence, cutting the voltage received by the non-inverting input of the integrator’s op-amp to half of the actual input voltage. This voltage drop forces the inverting input to match the non-inverting voltage, causing the integrator output to ramp down. The ramped down output hits the lower threshold set by the Schmitt Trigger and causes its output to snap low (-5 V). This output level turns off the MOSFET and forms a loop which produces a continual triangular and pulse wave.

The IRFZ44N MOSFET proved suitable for this purpose due to its minimal on-state resistance of 17.5 m Ω , allowing for instantaneous discharge of the capacitor upon

System Design Methodology

receipt of the pulse signal. This specification keeps the structure of the waveforms uniform.

The resistor values of 51 k Ω formed an approximate 2:1 ratio with the input resistance of 100 k Ω to ensure the integrator circuit pulls about twice the current in the opposite direction when the MOSFET is on to create a uniform triangular wave. Appendix B details the oscillator linearity testing methodology.

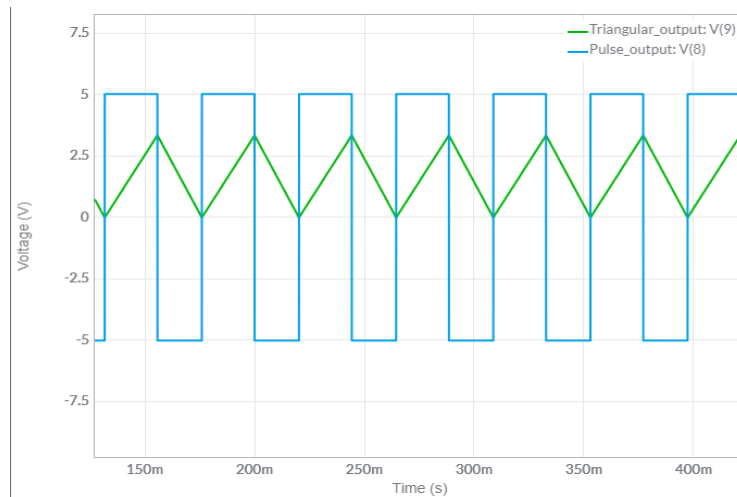


Figure 6. Pulse and Triangular Simulation Outputs of The Oscillator

2.4 Signal Combination

Carter and Brown (2001) outline the flexibility in the use of the basic inverting amplifier configuration for summing applications. The functions are easily varied based on parameter values hence providing a suitable structure for the mixing stage.

The system incorporates inverting op-amp configured summing amplifiers to combine the distinct waveforms produced by the two oscillators to produce a single harmonically enriched signal. The design takes on a two-stage mixing approach, dividing the signals into a sub-mix stage and an overall output stage.

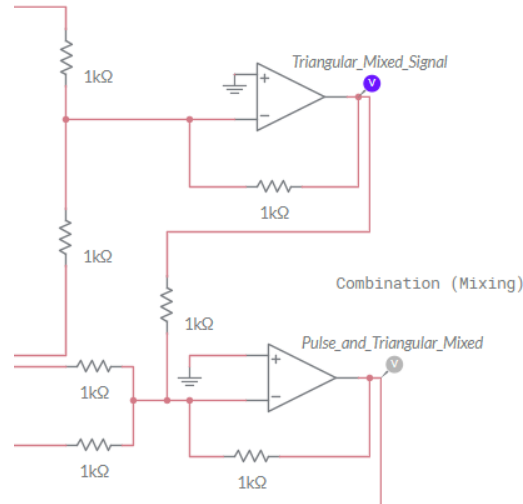


Figure 7. Signal Combination Stage Schematic

The first stage of the signal combination operates for the two triangular waveforms produced by the integrators on the oscillators. The subsequent mixing stage produces the final combined output by combining the pulse waves from the Schmitt trigger outputs on the oscillators with the previously combined triangular waves.

The two-stage mixing structure opens an avenue for control. The triangular waves are smooth compared to the sharp and harsh pulse waves. The triangular waves can be utilised as a tool to provide the sound signal with a ‘bass’ sound. Combining the triangular waves separately allows for bass control on the sound signal by varying the gain of the summing amplifier of the first stage. To achieve this control effect, a potentiometer replaces the amplifier’s feedback resistor.

$$Gain = -\frac{R_{feedback}}{R_{inputs}} \quad (2.2)$$

The one kilo ohm resistors in the schematic provide a resistance low enough to keep thermal noise negligible, and high enough to regulate the current directed to the mixer.

2.5 Filtering Stage

The system implements the second-order Sallen-Key low-pass filter as defined in the literature from the previous chapter(Kladovščikov et al., 2019). This specific structure of filter entails advantageous properties such as minimal component count and high-standard electrical stability.

The mathematical equation governing the cut-off frequency of the filter derives from the transfer function in equation 1.1.

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$$\omega_0 = 2\pi f_c = \frac{1}{\sqrt{C_1 C_2 R_1 R_2}} \quad (2.3)$$

$$\therefore f_c = \frac{1}{2\pi\sqrt{C_1 C_2 R_1 R_2}} \quad (2.4)$$

The design uses identical pairs of 5.6 nF capacitors and 33 kΩ resistors to establish a cut-off frequency of 861.23 Hz. The decision to use this value allows the filter to preserve the fundamental pitches of most synthesizer melodies which range from 50 Hz to 800 Hz, while removing excessive ‘buzz’ effects on the sound caused by higher pulse harmonics. Figure C.1 in Appendix C contains the circuit diagram of the filter used.

2.6 Attenuation Stage

The system implements an inverting amplifier structure for attenuation at the end of the analogue section. This architecture stands ahead of its non-inverting counterpart as it is more versatile in both increasing and reducing the amplitude of the signal as opposed to the non-inverting nomenclature only being able to increase the amplitude. The non-inverting amplifier’s gain equation shows evidence of this problem.

$$Gain = 1 + \frac{R_{feedback}}{R_{input}} \quad (2.5)$$

The inverting amplifier’s gain is determined by the equation 2.2. A potentiometer knob replacing the feedback resistor controls the gain value, hence the signal’s volume. The increase and reduction of the gain value have a directly proportional effect on the amplitude of the filtered waveform without altering the frequency of the signal. Figure C.2 and Figure C.3 in Appendix C represent the attenuation stage circuit diagram and the full analogue circuit schematic, respectively.

2.7 Microcontroller Implementation

This segment involved the use of an Arduino microcontroller to regulate DC voltage levels into the signal generation stage through a digital potentiometer to autonomously vary the frequencies and play a melody. An Arduino proved suitable for this purpose due to its high flexibility, seamless integration capabilities, and simple programming. The digital potentiometer chosen was the MCP4201. This device's SPI protocol allows the potentiometer to receive rapid and synchronised digital control signals from the Arduino. The 8-bit operational resolution as defined in its datasheet provides 256 distinct steps for voltage division hence producing highly precise inputs. The MCP4201 is also a dual chip which aids in component reduction considering the dual-core structure of the synthesizer.

The Arduino's programming used its IDE in the C language and connected to the synthesizer oscillators via the digital potentiometer. Appendix E outlines the C code use to program the Arduino.

The formulas below calculate the expected responses based on the digital inputs.

1. The target voltage depending on target frequency.

$$V_{target} = \frac{f_{target} - f_{offset}}{f_{pervolt}} \quad (2.7.1)$$

Where f_{offset} is the frequency at zero volts and $f_{pervolt}$ is the frequency at one volt.

2. Target frequency of the n^{th} piano key. (Iowa State University)

$$f_{target} = 440 \times 2^{\frac{n-49}{12}} \quad (2.7.2)$$

3. The potentiometer step based on the target voltage.

$$Step = \left(\frac{V_{target}}{V_{max}} \right) \times Step_{max} \quad (2.7.3)$$

Where $step_{max}$ is 256 and V_{max} is the voltage supply to the potentiometer. The value is rounded to the nearest integer since the digipot steps are distinct.

4. The actual voltage the step allows to pass through.

$$V_{actual} = \left(\frac{Step}{255} \right) \times V_{max} \quad (2.7.4)$$

5. The theoretical output oscillator frequency

$$f_{theoretical} = (V_{actual} \times f_{pervolt}) + f_{offset} \quad (2.7.5)$$

Chapter 3: Physical Implementation and Testing

The focus of this chapter is the physical construction of the dual-core music synthesizer based on the theoretical design and outlining the testing methods used to verify its functionality.

3.1 Testing Equipment

The main instrument used for extracting scientific data from the system was the KEYSIGHT NO-DSOX1102G oscilloscope provided by the laboratory. This oscilloscope provides a rapid sampling rate accurately plotting waveforms from the outputs of the separate functional subsystems. The oscilloscope exports experimental data, to the accuracy of every microsecond, in a .csv spreadsheet format via a USB port.

3.2 Power Supply

A bench power supply, also supplied by the laboratory, powered the analogue components of the system. This equipment DC voltages within a range of +20 volts to -20 volts from two output ports. Other ports include a ground terminal and a separate DC voltage output producing two distinct values of 3.3 volts and five volts only.

3.3 Prototype Construction

The final construction incorporated the distinct functional blocks on separate, smaller breadboards to aid in visual demonstration and testing of individual subsystems. The construction layout strictly followed the architectural sequence outlined in the previous chapter. Solid-core hook-up wires directed the audio signals through the system. This wire shows properties of high malleability as well as compatibility with high-frequency applications. A multimeter displayed resistance values between connections probed for continuity tests to ensure appropriate signal flow and power supply. Voltage levels of ± 15 V supplied the positive and negative terminals of all the op-amps due to the available bench output ports. Potential dividers were implemented at the triangular and pulse outputs of the oscillators to produce results likened to the simulations using ± 5 V

Physical Implementation and Testing

rails instead. To obtain this three times reduction, the equation below determined the resistor values.

$$V_{out} = \frac{R_2}{R_1 + R_2} \cdot V_{in} \quad (3.1)$$

A pair of one kilo ohm and two kilo ohm resistors sufficed.

The third bench output supplied a voltage of five volts to the digital potentiometer giving a DC voltage range of zero volts to five volts for the Arduino microcontroller to sequence through. Appendix F details the pin connections used during the prototype construction.

3.4 Troubleshooting

The initial physical implementation of the system did not function as expected at the first instance. The system's output did not show the expected audio signal waveform, rather it produced a slightly deformed straight, noisy signal line. Appendix D and the analysis in Chapter 4, outlines the system's expected output waveforms.

To pinpoint where the issue was located, each functional block was isolated and evaluated using the oscilloscope to determine whether its output signal was of an expected waveform. After each subsystem went through this evaluation, the next subsystem was connected, and the whole mini system underwent evaluation. This iterative process continued until the oscilloscope displayed an unexpected waveform at the output of the signal combination subsystem. Instead of the expected harmonically dense waveform, the oscilloscope showed a straight signal line similar to the output of the system during the initial test.

Further multimeter continuity tests discovered a misconnection between the pins of the op-amps in the signal combination section and their expected inputs. The multimeter test between the negative voltage supply and the negative voltage pin of the first mixing stage's op-amp displayed a 'one,' without the beeping sound which normally indicates a connection between two points.

The wiring was corrected based on these findings and the system was put back together. A final oscilloscope test showed the prototype functioned as expected.

Chapter 4: Results and Analysis

This chapter presents and analyses the data extrapolated from the music synthesizer's physical implementation. Graphical presentation showed the signal progression through the system, confirming the functionality of each functional stage of the system and their impact on the audio signal.

4.1 Signal Generation Analysis

The VCO design expects the signal generation stage to produce triangular and pulse waveform outputs from the integrator and Schmitt trigger outputs respectively (Y. Xu et al., 2021).

A five-volt DC signal drove the VCOs to produce the baseline test frequencies. The horizontal, one millisecond per division oscilloscope setting captured a suitable number of waveform periods.

4.1.1 Integrator Outputs

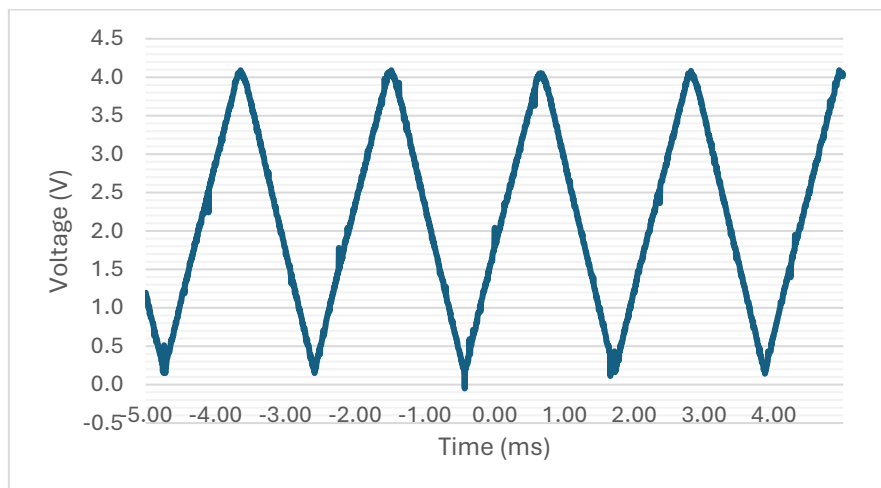


Figure 8. Oscilloscope graph of the triangular output from VCO1

The maximum voltage peak represented in the graph was four volts and the minimum was zero volts, defining a peak-to-peak voltage of four volts. The oscilloscope measured a 463.5 Hz frequency for this wave.

Results and Analysis

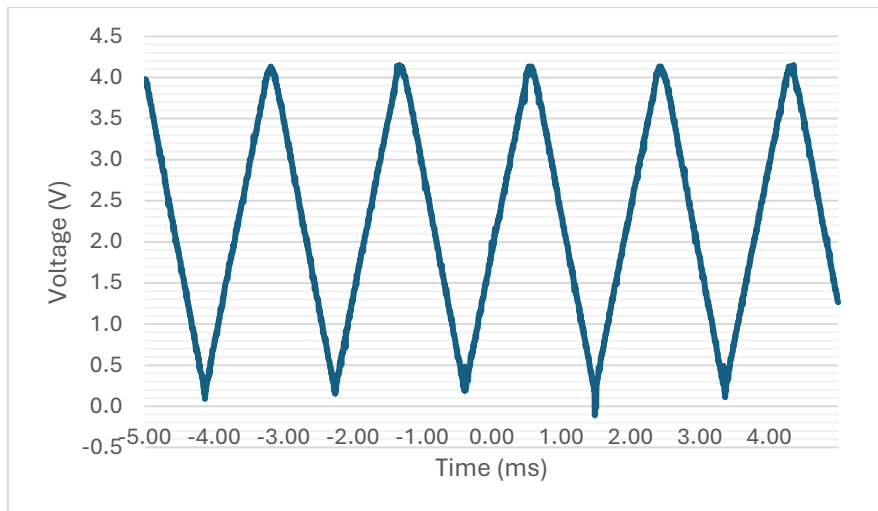


Figure 9. Oscilloscope graph of the triangular output from VCO2

The second triangular waveform was similar to the first in terms of the peak-to-peak voltage values and presence in the positive voltage domain. However, measurement readings revealed a notable frequency difference between the two waveforms from identically designed circuits. VCO2 operated at a frequency of 535.6 Hz.

This 72.1 Hz frequency difference represents a 14.4% deviation. This variance stems from physical component tolerances within the resistors and capacitor at the integrator. The 2.2 nF ceramic capacitors used for construction possess a manufacturing tolerance of $\pm 10\%$, while the resistors possess a $\pm 1\%$ range. Since the oscillation frequency is inversely proportional to capacitance ($f \propto 1/C$), variance in capacitor values would explain this frequency discrepancy. The pulse outputs from the Schmitt trigger would also demonstrate the same variance since the triangular waves from the integrator drives them. This effect opens up an avenue to analyse signal combination involving waveforms of different frequencies and to further demonstrate to students the effects of component tolerances in pitch accuracy.

Both figures above demonstrate expected waveform shapes. They clearly show the ramp-up and ramp-down effects caused by the integrator as a consequence of voltage changes caused by the MOSFET switching action.

Results and Analysis

4.1.2 Schmitt Trigger Outputs

Since the architecture of the VCOs requires the triangular integrator outputs to drive the Schmitt trigger, the pulse waveforms produced were at the same frequencies as their triangular wave counterparts.

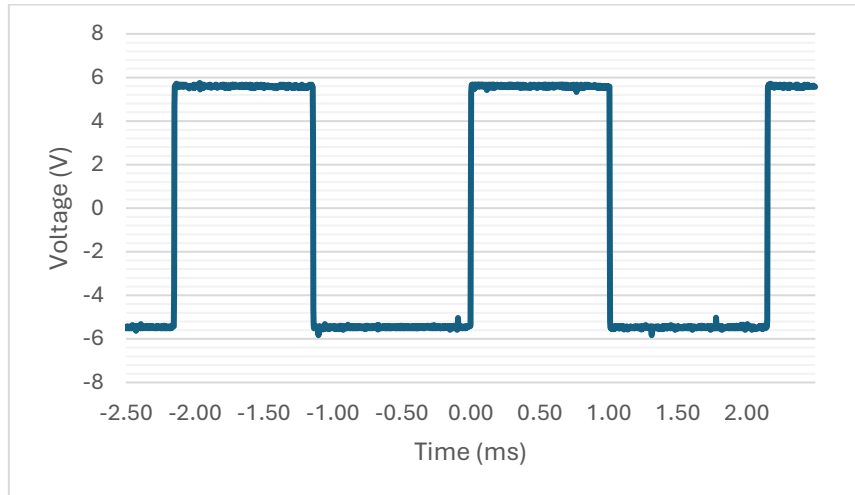


Figure 10. Oscilloscope Graph of the Pulse Output from VCO1

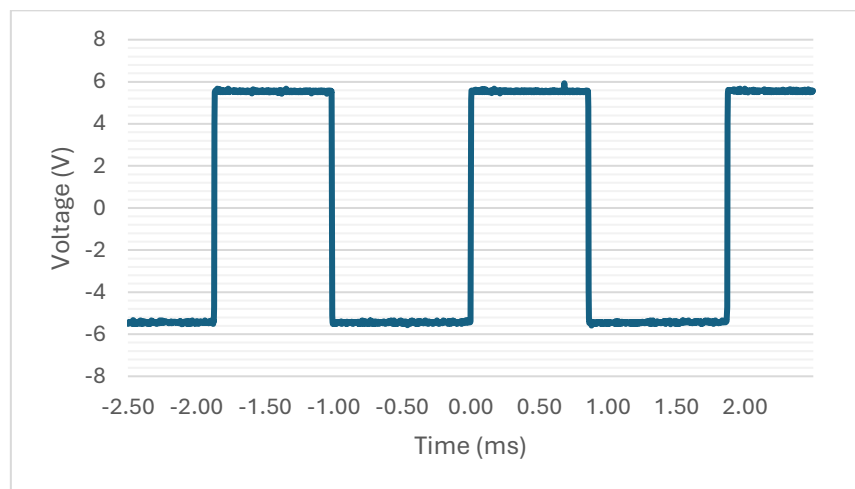


Figure 11. Oscilloscope Graph of the Pulse output from VCO2

The Schmitt triggers efficiently converted the triangular voltage ramps from the integrator into pulse waves. The graphs show the sharp transitions before the voltage settles at +5.5 V and -5.5 V. This sharp transition effect shows the advantageous impact of the TL082CN op-amp's 13 V/ μ s slew rate. Comparative analysis with the software simulation results in figure 6 shows a high degree of waveform shape correlation, proving the physical implementations of the oscillators function correctly.

Results and Analysis

4.2 Signal Combination Analysis

The following graphical results demonstrate the functionality of the two summing amplifiers in combining the signals from the oscillators together. The exact signals shown in the previous results progressed through the mixers to produce these graphs. A five millisecond per division horizontal oscilloscope setting captured the full distortions on the resulting waveforms.

4.2.1 First Mixing Stage

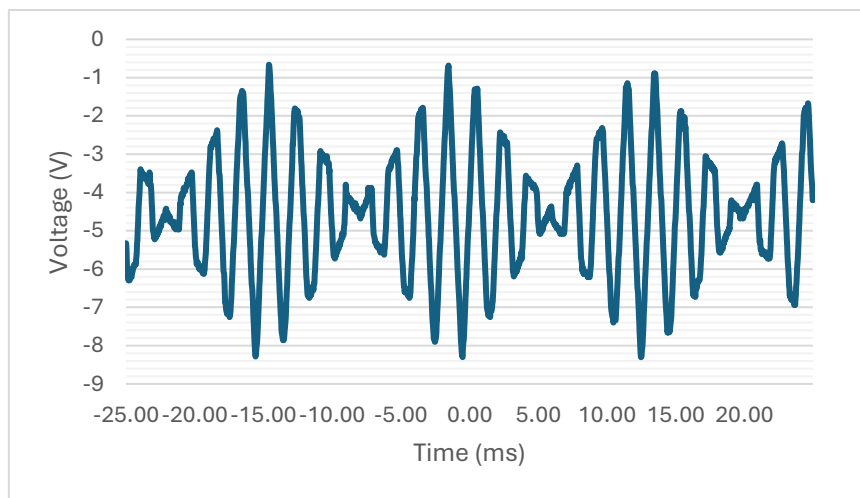


Figure 12. Oscilloscope Graph of the First Mixer Output

This mixer combined the triangular waves from both VCOs. The resulting waveform demonstrates signal heterodyning, a principle outlined by the National Institute of Standards and Technology (2008) for analysing the combination of signals of different frequencies. The phases of the oscillator outputs stray in and out of alignment, changing the amplitude over time via constructive and destructive interference, and produces an effect known as beat frequency.

Due to the ± 15 V rail supply to the op-amps, no voltage clipping occurred at this stage. The maximum voltage peak was -0.7 V, and the minimum was -8.4 V, giving a maximum peak-to-peak voltage of 7.7 V. The voltage levels are in the negative region due to the inverting configuration of the summing amplifier.

Results and Analysis

4.2.2 Second Mixing Stage

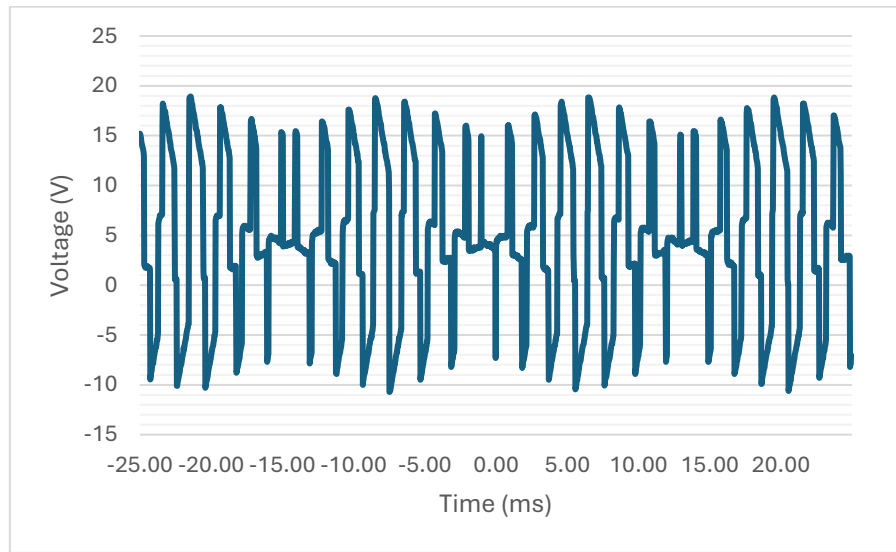


Figure 13. Oscilloscope Graph of the Second Mixer Output

The second mixing stage combined the output of the first stage with the two pulse outputs from the VCOs into the waveform shown above. The Schmitt trigger pulse outputs from the oscillators drove the massive, rapid vertical jumps shown in figure 13. The second mixer operates similarly with the triangular waves by combining amplitudes when the signals are in perfect alignment and obtaining smaller peaks during periods of misalignment.

The maximum voltage peak was +19 V, and the minimum was -11 V. Hence, the maximum peak-to-peak voltage was 30 V, consistent with the ± 15 V rails provided to the op-amp. The fundamental frequency of the waveform was determined as follows:

$$T = \frac{\text{Time Sample}}{\text{Cycles}} \approx \frac{10 \text{ ms}}{5} = 2 \text{ ms} \quad (4.1)$$

$$f = \frac{1}{T} = \frac{1}{0.002} = 500 \text{ Hz} \quad (4.2)$$

This frequency value is consistent with the average of the two VCO frequencies (499.6 Hz).

The same signal heterodyning effect from the previous stage is present in this stage. The ‘shrinking’ effect of the waveform occurred at periods of 14 ms, which is the beat period. The inverse of this period determined the difference in frequency of the two oscillators.

$$\Delta f = \frac{1}{T_{\text{beat}}} = \frac{1}{0.014} \approx 71.43 \text{ Hz} \quad (4.3)$$

Results and Analysis

This value almost perfectly aligns with the measured difference in frequency of 72.1 Hz with an error margin of only 0.93%.

The tops or plateaus of the waveform slant. This effect occurred from layering the mixed triangular waves from the first mixer with the combined pulse waves. The triangular wave ‘rides’ on the back of the rapid vertical ‘boosts’ caused by the pulse waves, causing the plateaus to smoothly ramp up or down until another aggressive snap. Appendix D explores the results of signal combination of equal frequency waveforms.

4.3 Filter Function Analysis

The same oscilloscope setting from the previous stage captured the waveform below.

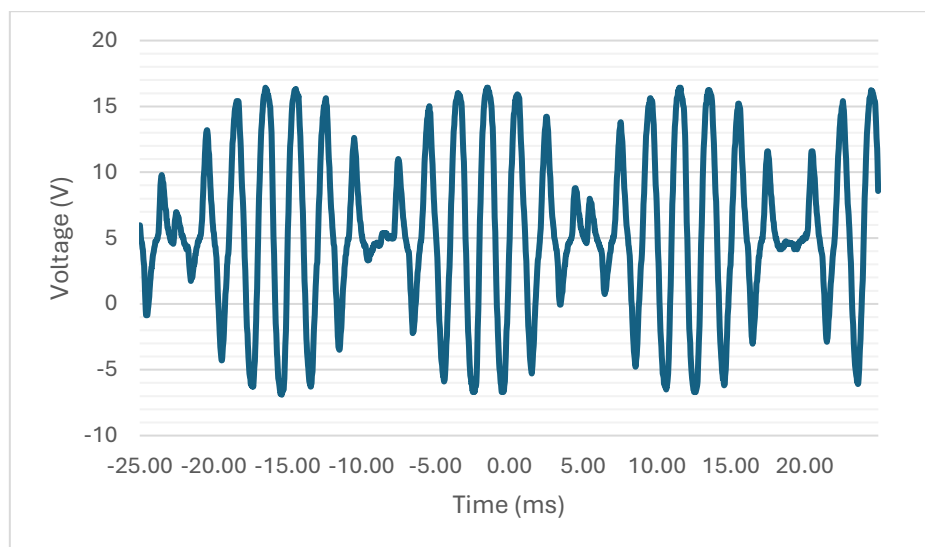


Figure 14. Oscilloscope Graph of the Filtered Combined Signal

Figure 14 above demonstrates the ‘smoothing’ effect of the second order Sallen-Key filter on the combined signal in figure 13. The filter effectively attenuated the high frequency energy hence, the rounding effect at the peaks. The rounding effect from the filter causes the maximum peak-to-peak value of this waveform to reduce from 30 V to 24.5 V. The fundamental frequency of the waveform remained unchanged from the unfiltered signal, however, there is a delay induced causing a slight phase shift of five milliseconds in the output of the filter compared to the unfiltered signal.

Results and Analysis

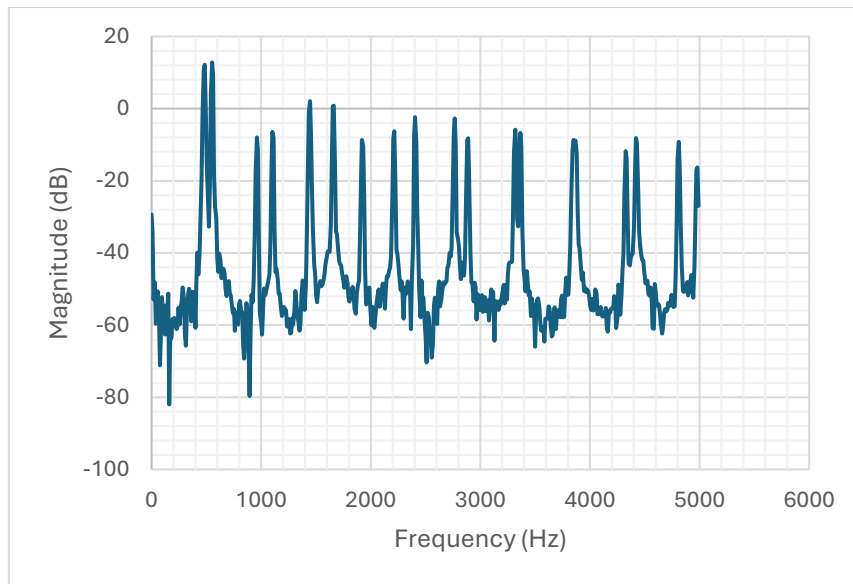


Figure 15. Fast Fourier Transform (FFT) Analysis of the Unfiltered Combined Signal

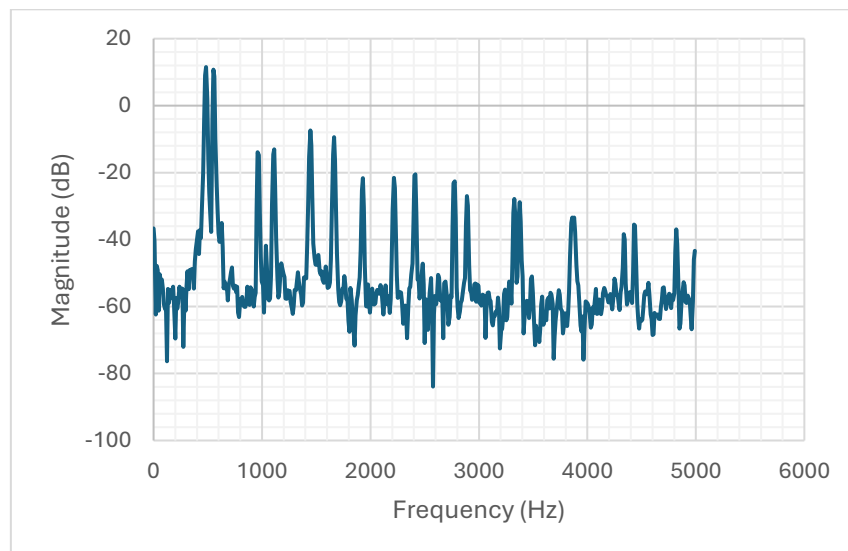


Figure 16. Fast Fourier Transform Analysis of the Filtered Combined Signal

The oscilloscope captured the FFT data using a configuration of a 10 milliseconds per division time base to capture a 100 millisecond sample window. A 500 hertz per division horizontal scale displayed the cut-off frequency point and two octaves above it, to clearly indicate the effects of the filter at certain frequencies.

Calculations in the design phase determined that the 5.6 nF capacitors paired with the 33 k Ω resistors set a specified cutoff frequency of 861.2 Hz.

Results and Analysis

Sallen & Key (1955) outline that the roll-off rate of an active filter follows a $-6n$ dB/octave, with n being the filter order. Since this a second-order filter, there is an expected attenuation of -12 dB per octave after the cut-off frequency point.

The FFT analysis pre and post filtering, proves this theoretical calculation. In the graph produced from the unfiltered signal in figure 4.8, there was little attenuation shown in the harmonic peaks across the frequency axis. The magnitudes of the harmonics sat mostly between 0 dB and -10 dB.

Figure 16 shows the effect of the filter on the combined signal. The fundamental core frequencies at 500 Hz are well below the 861.2 Hz cutoff hence remaining completely unaffected. The higher frequency harmonics, however, degraded following the -12 dB/octave theoretical rule.

Table 3. Comparison of FFT Magnitude between the Unfiltered and Filtered Signal

Octave above f_c	Frequency (Hz)	Unfiltered FFT Magnitude (dB)	Filtered FFT Magnitude (dB)	Magnitude Difference (dB)
1 st	1722	4	-8	-12
2 nd	3444	-4	-28	-24

These values present magnitude drops of 12 dB and 24 dB respectively, perfectly aligning with the theoretical expectations.

Results and Analysis

4.4 Output Attenuation

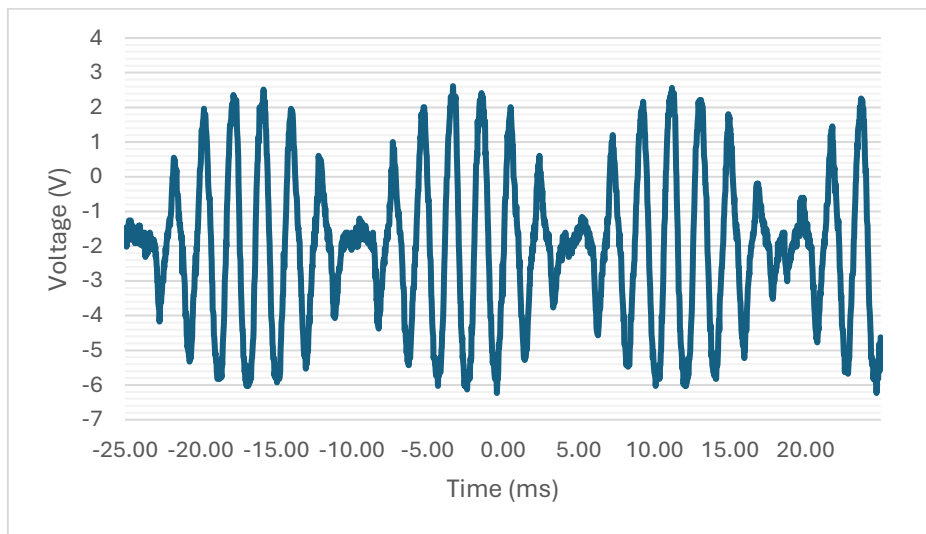


Figure 17. Oscilloscope Graph of the Attenuated Filtered Signal

Figure 17 shows the output waveform after the attenuation stage. The main differences observed compared to figure 14 were the inversed topology of the waveform, and its reduced amplitude. The adoption of an inverting amplifier model accounts for the first effect. The latter was due to the gain level of the inverting amplifier which is variable depending on the level of resistance set by the feedback potentiometer. In the case shown above, the potentiometer was set to a value of $5.2\ \Omega$, coupled with an input resistance of $15\ \Omega$ to achieve a gain value of -0.347 . This gain value attenuated the peak-to-peak voltage of the filtered waveform from $24.5\ \text{V}$ to $8.5\ \text{V}$. The frequency of the signal, however, remained unchanged after attenuation.

Results and Analysis

4.5 Digital Control Response

The code detailed in Appendix E programs the digital control interface for the analogue system. Comparison of the theoretical frequency calculations using the equations from 2.7, and the measured output frequency from the first VCO took place to evaluate the subsystem's functionality.

Table 4. Digital Control Accuracy for VCO1

Target Note	Target Frequency (Hz)	Calculated Digipot Step	Calculated Frequency (Hz)	Measured Frequency (Hz)	Error against target (Hz)
C	261.6	132	260.8	260.6	-1
D	293.7	150	293.5	293.8	0.1
E	329.6	170	329.9	330.6	0.4
G	392.0	204	391.7	391.5	-0.5

The table above demonstrates the predictability of the system based on the digital inputs. The error margin of the measured frequency compared to the target frequency is within a ± 1 Hz range. This slight frequency drift results from the quantisation limits present in the digital potentiometer which round the required voltage to the nearest available step.

Chapter 5: Discussions and Conclusions

This chapter evaluates the overall effectiveness in the project's ability to meet the initial aims and objectives outlined in the first chapter. The following discussion entails the educational incorporation of the system, challenges faced in the construction of the system, and openings for potential improvement in the future.

5.1 Evaluation against Project Aims

The main engineering goal of the project was to create an interactive medium to practically demonstrate complex analogue electronics theoretical concepts. The project provided a solution to this suggestion made by Pagiatakis and Voudoukis (2017), with the theoretical design and physical implementation of a dual-core music synthesizer.

The oscilloscope testing methodology was able to visually prove how each of the subsystems functioned in manipulating the DC voltage signal into an enriched audio signal. Students could observe the oscilloscope waveforms at a particular setting, input changes through the Arduino, and note the real-time changes incurred on the signal. Quantitative analysis of the results validated mathematical equations governing analogue electronics operations including the functionality of op-amps.

The project satisfies the outlined technical objectives in the following ways. The implementation and simulation of the architecture in NI Multisim defined the functional system structure and observed its expected behaviour. The MCP4151 digipot driven by the Arduino code in appendix E, achieves the third objective of developing a digital control interface. Physical implementation and data extraction from the system via the oscilloscope, satisfied the construction and evaluation objectives.

5.2 Educational Incorporation

Similar to the eight-week model outlined by Billis et al (2011), this synthesizer design should form a semester-long, practical laboratory module within the final year of the BEng Electronic and Electrical Engineering degree. The extended duration of this course would give students the opportunity to understand in depth, the physical operation and theoretical background of each functional stage/component one at a time. The final year of the BEng degree is a suitable placement for this module because students

Discussions and Conclusions

would have studied or be in the process of studying the more advanced electronic concepts present in the system such as active filter operation. The school laboratory would provide the necessary equipment for design, construction, and testing of the system.

However, lab-time availability may not be sufficient to complete all work at school. To counter this issue, the school should provide lab equipment kits to student groups assigned for this module. The kits would include all circuit components required for the synthesizer design as well as a small power module, LEDs, and a speaker for testing outside the laboratory. The power module should be a battery holder case box fitted with breadboard compatible wires. The lab kit should include two 15 V lithium batteries to mirror the power source from the bench in the lab. Students should fit LEDs to the outputs of the functional stages to give a rough estimate of the signal's nature i.e. LEDs should flicker at a faster frequency with higher input voltage to the VCO. The students should also fit the speaker provided at similar points to audibly detect changes in pitch compared to the power supply.

Teachers should assess students at the end of the module with a presentation of the functionality of the overall system and each subsystem. The assessment should also cover circuit structure and additional control elements.

Common student errors lab assistants can look out for over the course of the module include misreading of resistor colour bands and improper rail supply to op-amps.

5.3 Challenges Faced

Despite the successful physical implementation of the design, the project faced specific limitations. The main challenge encountered was the relation of the results obtained from the system back into an educational environment. This problem arose due to the relative lack of dense academic material on the use of music synthesizers as an educational tool for teaching electronic concepts as compared to traditional engineering topics.

Discussions and Conclusions

5.4 Future Work

Future development on this project could entail an expansion of the digital control section to increase the interaction level of the system. This expansion would include an extra set of digital potentiometers for Arduino control using refined code on other functional subsystems such as the filtering and signal combination stages.

Future work could also involve developing additional C programs of different tunes and melodies to enable students to completely change the outputs of the music synthesizer with minimal effort.

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Appendices

Appendix A: Project Management

task	wk1	wk2	wk3	wk4	wk5	wk6	wk7	wk8	wk9	wk10	wk11	wk12	Milestones	Deliverables
1													Conduct extensive research	
2													Identify components	Individual structure simulations
3													Picking suitable structures	Graphical representation of simulation results
4													Test components isolated	Built circuit of an oscillator and analysis of waveforms
5													Compare results	Graphical representation of experimental results
6													Find suitable architecture	Simulation structure and graphs of full architecture
7													Prepare preliminary report	Preliminary report
8													Construct model	A functioning analogue system
9													Program Arduino	A functioning digital control unit
10													Create control inputs	Project presentation
11													Final report preparation	Final Report

Figure A.1 Project Gantt Chart

The Gantt chart above structured the general timeline of the project. This schedule adopted a sequential model, with tasks covering areas in research, prototyping, and physical implementation of the system.

Selection of Milestones and Deliverables

The project timeline was split into distinct phases to set a clear path towards its completion.

The first phase was defining a theoretical baseline for the project to form a reasonable starting point. Tasks one to three were related to research, structure identification, and simulations. These milestones led to the establishment of a potential system structure within a software environment and simulation results to serve as a theoretical reference.

The second phase validated the compatibility of the components within the system hence reducing risks in construction. Tasks four to six enabled me to discover how the components functioned individually and with each other. Of the tangible deliverables defined from these milestones was experimental data of physical implementations of individual functional subsystems, as well as simulation data of a full architecture of the subsystems operating in cohesion.

Appendices

The third phase implemented the design into a physical prototype. The largest number of weeks were allocated to this phase. The milestones involved the physical construction and control design of the final system. These milestones were set to obtain tangible experimental deliverables (oscilloscope data) proving the system's functionality.

Adherence to Plan

The project plan was strictly followed for the most part. However, some design choices were changed during the physical implementation phase.

The design of the oscillators used for signal generation changed from the initially evaluated version in phase two. This involved a change from the 10 nF capacitor in the integrator section to a 2.2 nF capacitor. This replacement increased the effective frequency range for the same zero-to-five-volt input voltage range.

This deviation took place simultaneously with the prototype implementation phase, allowing for system modification without affecting the overall timeline of the project.

Appendices

Appendix B: Oscillator Linearity Testing

To ensure effective tunability of the signal generation stage, an experiment performed on the oscillator architecture to determine its linearity.

The schematic used was the same as in Figure 5 in the main body.

Table B.1 Frequency Response Based on Input Voltage Magnitude

Voltage Supply (V)	Period (ms)	Frequency (Hz)
1	17.094	58.500
1.5	11.216	89.158
2	8.429	118.638
2.5	6.772	147.667
3	5.706	175.254
3.5	4.841	206.569
4	4.284	233.427
4.5	3.791	263.783
5	3.439	290.782
5.5	3.138	318.674

A graph plotted the data in the table above.

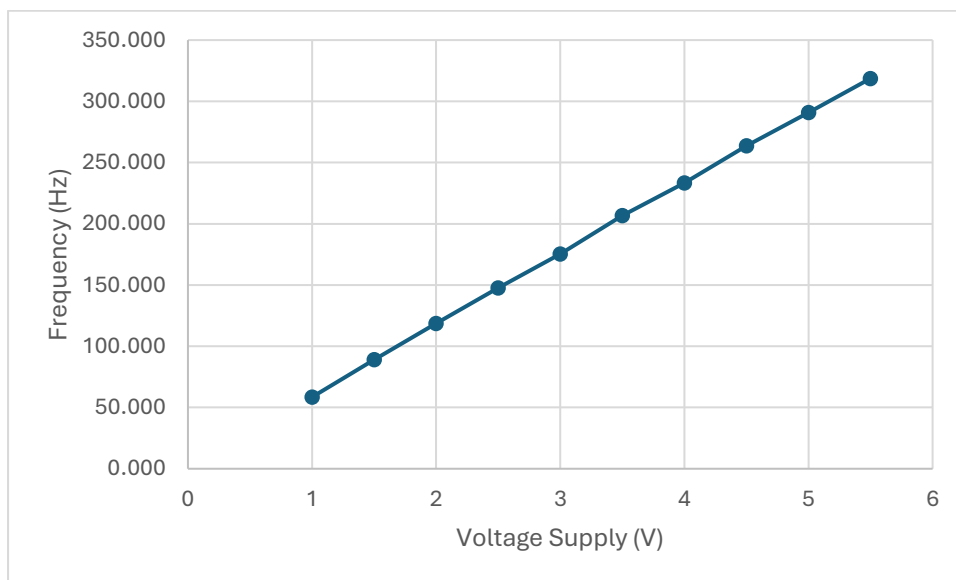


Figure B.1 VCO Frequency against Voltage Supply

The figure above shows that the output signal frequency of the designed VCO is directly proportional to the input voltage. This experiment further informed the choice on the VCO architecture due to its predictable behaviour.

Appendices

Appendix C: Circuit Schematics

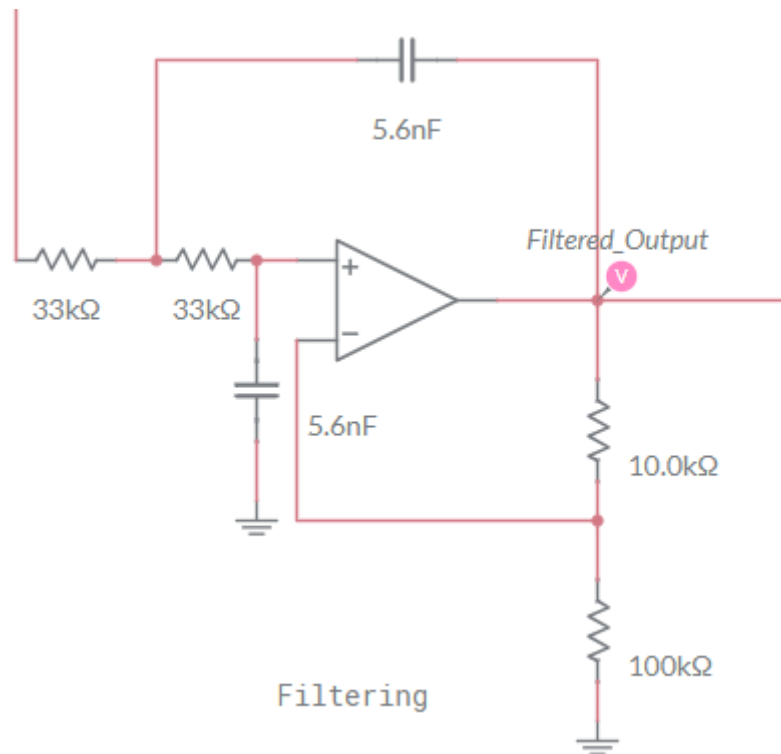


Figure C.1 Filtering Stage Schematic

Attenuation

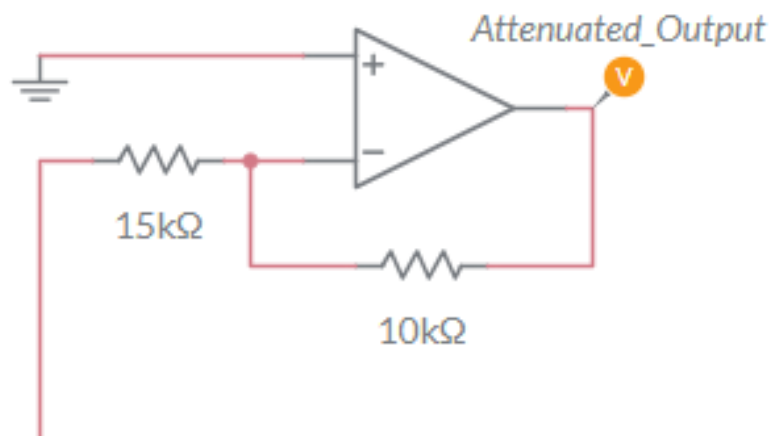


Figure C.2 Attenuation Stage Schematic

Appendices

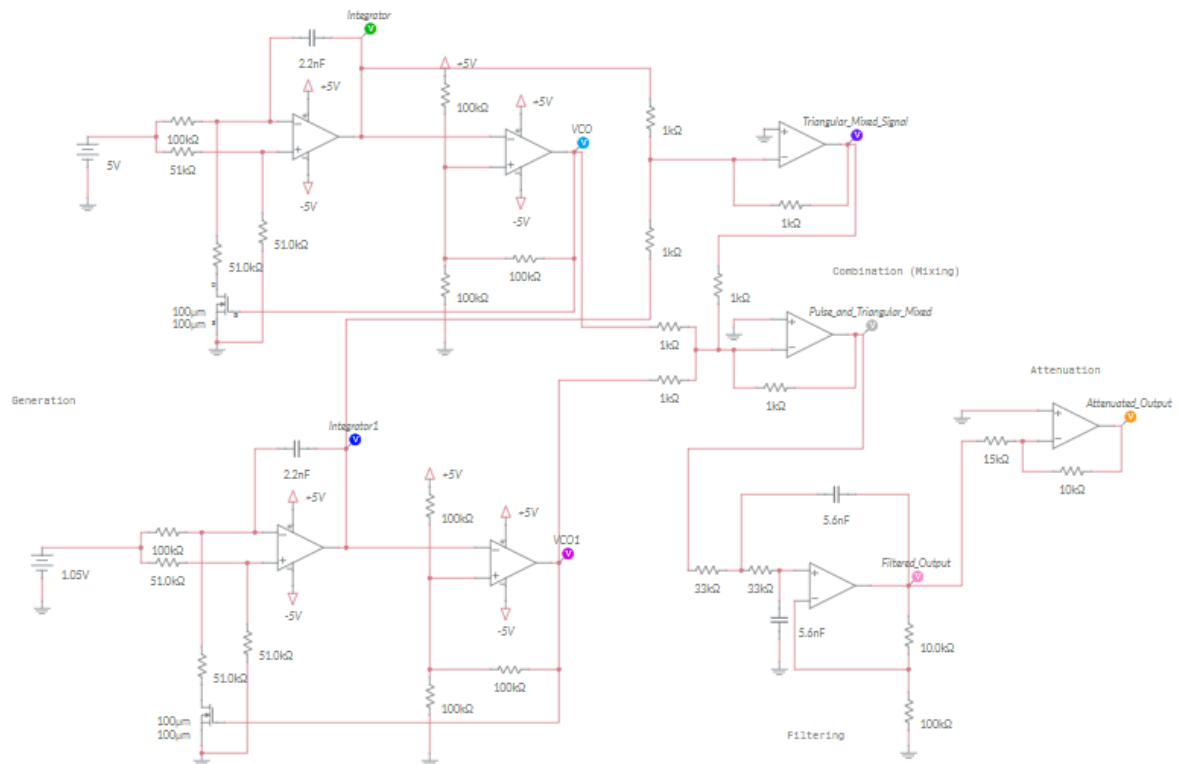


Figure C.3 Full Analogue Circuit Schematic

Appendices

Appendix D: Schematic Simulations

The following simulations demonstrate the functionality of the system subsystems when dealing with signals of the same frequency.

Signal Generation

The signal generated from the VCOs are consistent with the simulation results in figure 6 within the main body of the report.

Signal Combination

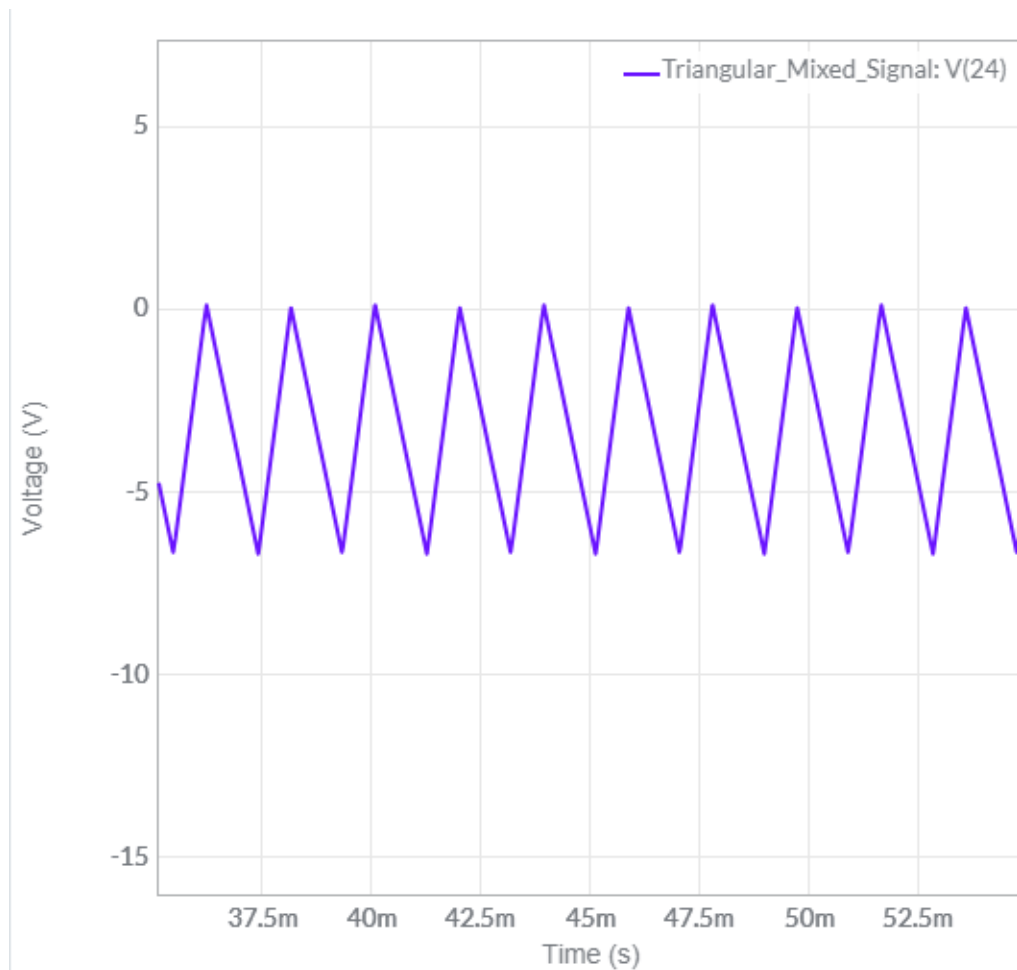


Figure D.1 First Mixer Output (Same Frequency Triangular Waveforms)

Appendices

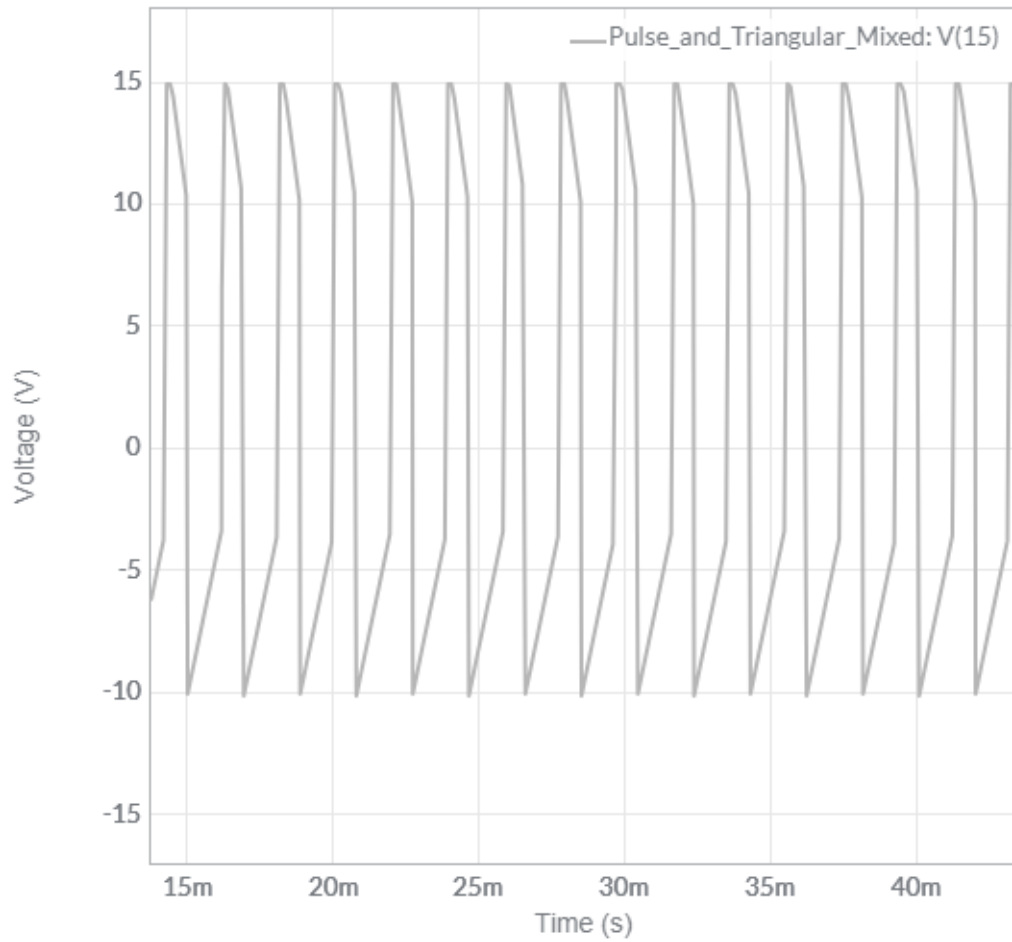


Figure D.2 Second Mixer Output (Same Frequency Pulse + Triangular Waveforms)

Appendices

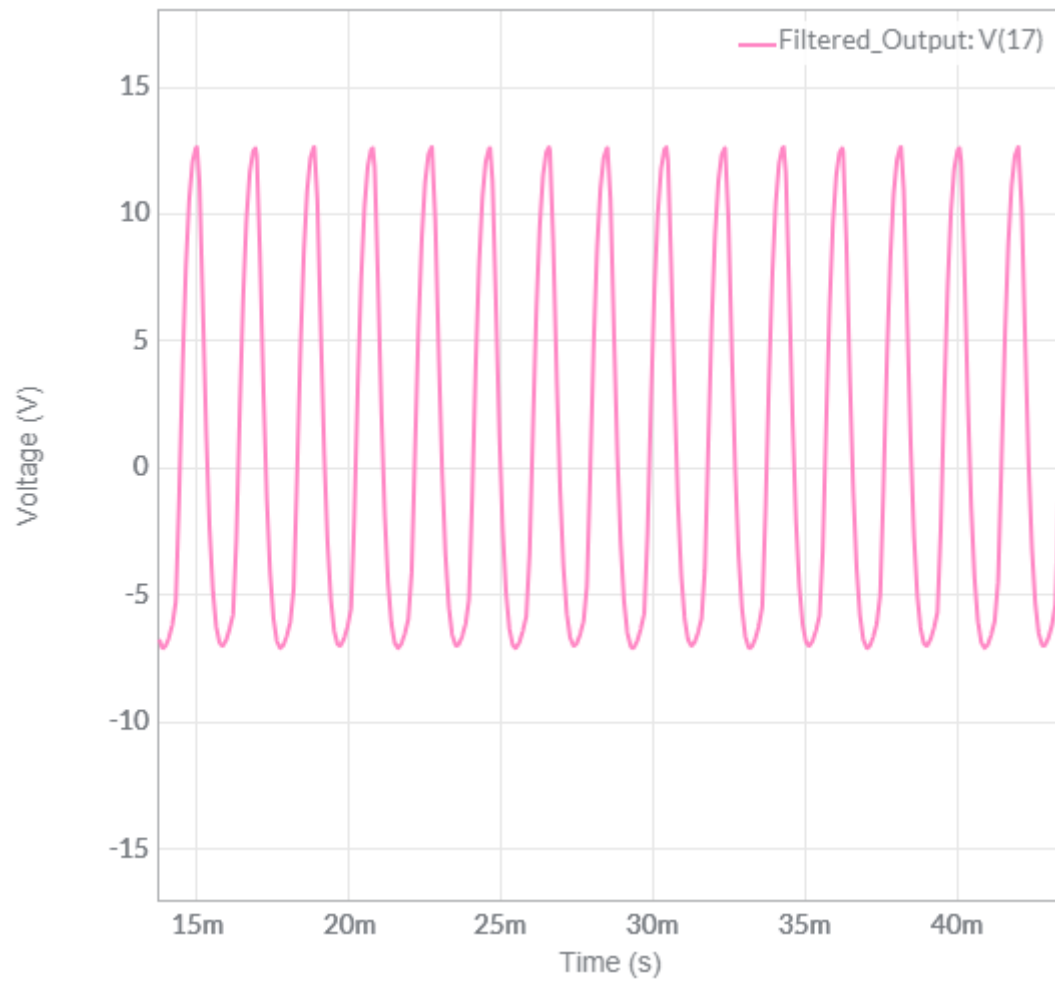


Figure D.3 Filtered Combined Signal

Appendices

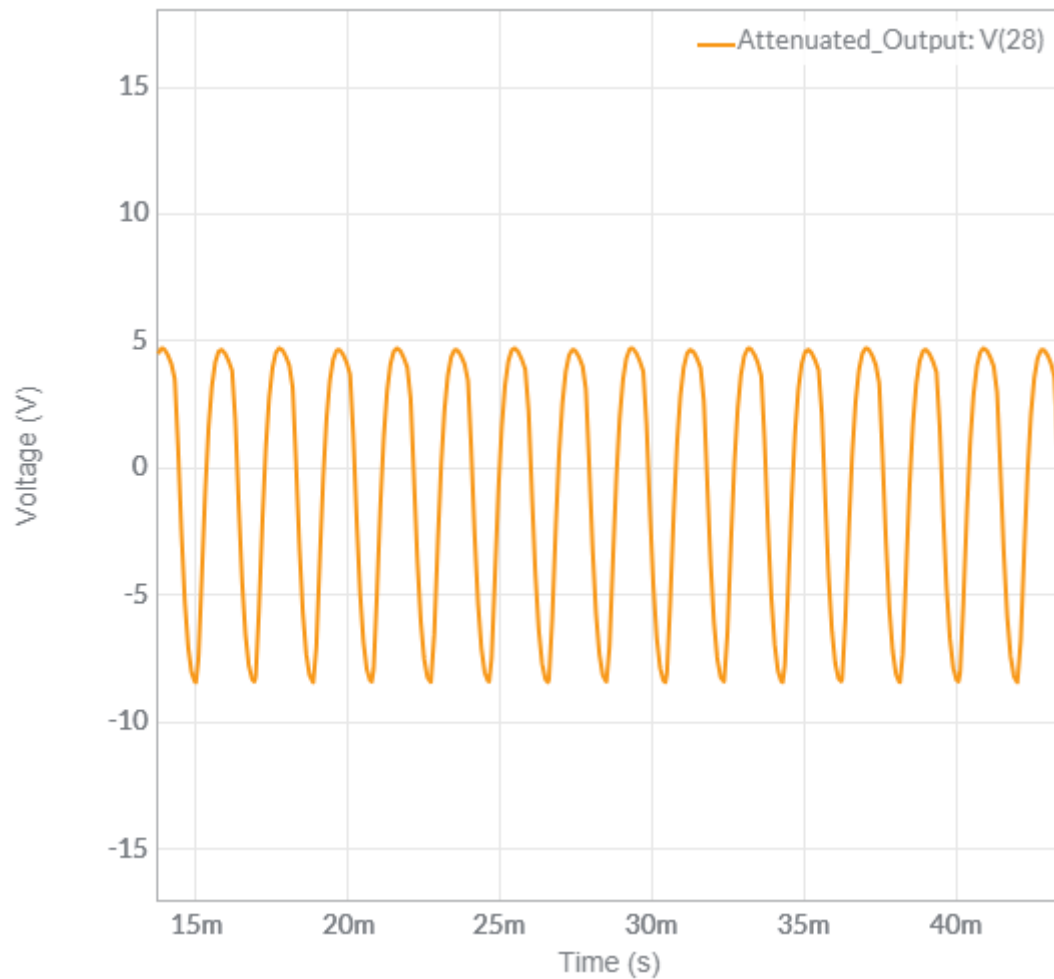


Figure D.4 Attenuated Filtered Signal

The most notable effect observed from the combination of signals with the same frequency against the physical experimental results is the absence of a beat frequency. Since the waveforms ramp up and down at the exact same times, there is always constructive interference hence the signals stack on each other. This led to the consistent doubling of the signal amplitude at every waveform peak. Aside from the absence of the beat frequency, every other effect on the signal observed during the experiments detailed within chapter 4, occurred in the simulation results.

Appendices

Appendix E: Arduino Code

The C code used to program the Arduino Microcontroller is shown below. The code controls the synthesizer to play *Mary Had a Little Lamb*, with the first VCO generating the melody, and the second VCO generating chord root notes.

```
#include <SPI.h>

// Configuration based on digipot datasheet information

const int CSPin = 10; //Chip Select Pin

const float max_voltage = 5.0; //Digipot Max Voltage

const int max_steps = 255; // 8-bit Digipot hence 256 distinct steps

// Frequency responses per volt of the two oscillators and voltage
offset

float vco1fpv = 92.700;

float vco1offset = 20.853;

float vco2fpv = 107.120;

float vco2offset = 25.894;

// defining note frequencies (Iowa State University)

const float C = 261.626, D = 293.665, E = 329.628, G = 391.995;
```

Appendices

```
// Notes for Bassline/Chords (Octave lower)

const float C_L = 130.81, G_L = 196.00;

const float rest = 0.0; // command to not play a melody note

const float hold = -1.0; // command to sustain bass notes


// Setting the melody notes for VCO 1

float melody1[] = {

    E, D, C, D, E, E, E, rest,

    D, D, D, rest, E, G, G, rest,

    E, D, C, D, E, E, E, E,

    D, D, E, D, C, rest

};


// Setting the bassline notes for VCO 2 (simple chords)

float melody2[] = {

    C_L, hold, hold, hold, hold, hold, hold, rest,

    G_L, hold, hold, rest, C_L, hold, hold, rest,

    C_L, hold, hold, hold, hold, hold, hold, hold,

    G_L, hold, hold, hold, C_L, rest
```

Appendices

```
};
```

```
// defining the note durations
```

```
float noteLengths[] = {
```

```
    1, 1, 1, 1, 1, 1, 2, 0.5,
```

```
    1, 1, 2, 0.5, 1, 1, 2, 0.5,
```

```
    1, 1, 1, 1, 1, 1, 1, 1,
```

```
    1, 1, 1, 1, 4, 1
```

```
};
```

```
// setting the tempo of the song
```

```
int temp = 250;
```

```
void setup() {
```

```
    pinMode(CSPin, OUTPUT);
```

```
    digitalWrite(CSPin, HIGH); // Programming Chip Select Pin
```

```
    SPI.begin(); // start SPI (Serial Peripheral Interface) bus
```

```
}
```

Appendices

```
void loop() {

    int Num_of_notes = sizeof(melody1) / sizeof(melody1[0]);

    for (int i = 0; i < Num_of_notes; i++) {

        // Melody control for VCO 1

        if (melody1[i] == rest) {

            pot_signal(0, 0); // silence it when 'rest' comes up

        } else {

            VC01_settings(melody1[i]);

        }

        // Bass/Chords control for VC02

        if (melody2[i] == rest) {

            pot_signal(1, 0);

        } else if (melody2[i] != hold) {

            // only update if it's not a hold

            VC02_settings(melody2[i]);

        }

    }

}
```

Appendices

```
// wait based on duration

int waitTime = noteLengths[i] * temp;

delay(waitTime);

}

delay(3000); // three second pause before looping again
}

// converting frequency to voltage for first VCO

void VCO1_settings(float freq) {

    float volts = (freq - vco10ffset) / vco1fpv;

    // determining the potentiometer step for each voltage value

    int step_value = (volts/max_voltage) * max_steps;
```

Appendices

```
    step_value = constrain(step_value, 0, 255);

    pot_signal(0, step_value);

}

// same thing for second oscillator

void VC02_settings(float freq) {

    float volts = (freq - vco2offset) / vco2fpv;

    int step_value = (volts / max_voltage) * max_steps;

    step_value = constrain(step_value, 0, 255);

    pot_signal(1, step_value);

}

// handles SPI writing to digipot
```

Appendices

```
void pot_signal(int pot_select, int val) {

    // choosing command depending on potentiometer index

    byte cmd;

    if (pot_select == 0) {

        cmd = 0x11;

    } else {

        cmd = 0x12;

    }

    // Chip select logic for control of each potentiometer

    digitalWrite(CSPin, LOW);

    SPI.transfer(cmd);

    SPI.transfer(val);

    digitalWrite(CSPin, HIGH);

}
```

Appendices

Appendix F: Physical Circuit Connections

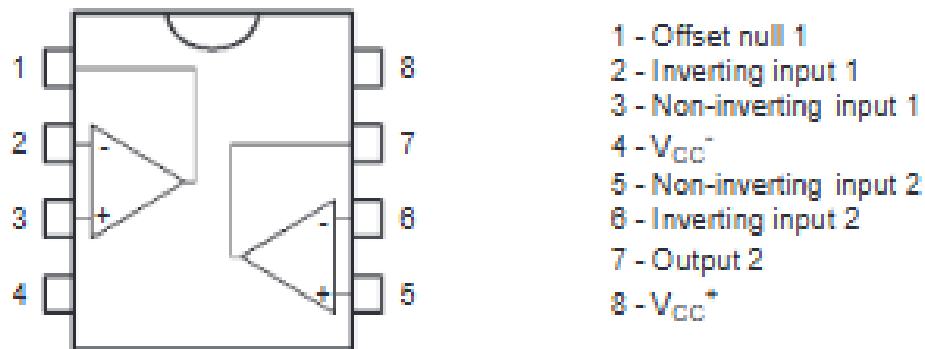


Figure F.1 TL082CN Pin Connections (STMicroelectronics, 2001)

The above figure demonstrates the pin locations of the TL082CN op-amp used in all the system's functional blocks. The ± 15 V rails connect to pin eight and four of each op-amp using red and orange wires, respectively. White wires routed signals in between functional stages, and black wires represented ground connections. All connections to specific op-amp pins remained consistent with the schematic in figure C.3.

Each VCO made use of one single chip as there are two op-amps inside. The integrator employed the use of the first op-amp (left), and the Schmitt trigger used the second (right). The mixers made use of one chip each due to space provisions on the breadboard. The filtering and attenuation stages also used one chip to separate functional stages on different breadboards. A potentiometer replaced the 10 k Ω feedback resistor in the attenuation stage to provide manual volume control.

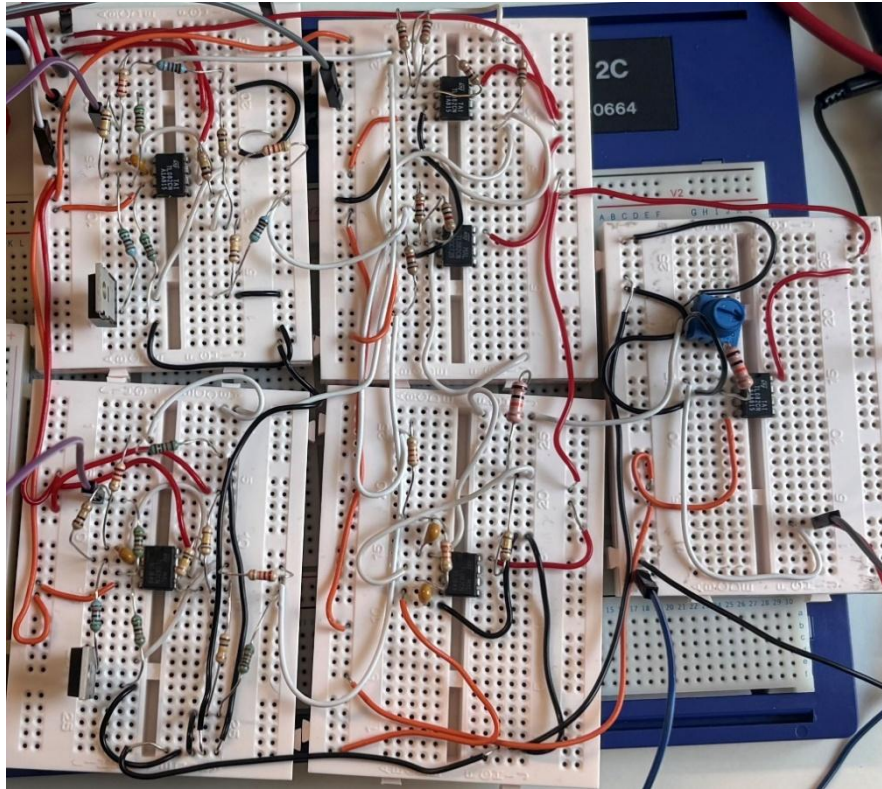


Figure F.2 Physical Circuit Implementation

Appendix G: Risk & Ethics Assessment

Electronic, Electrical and Systems Engineering Risk assessment for generically assessed operation

- ✓ Using equipment that has been purchased and provided by the School
- ✓ Using equipment that has a valid tested for electrical safety label
- ✓ Using a bench power supply that provides less than 30 Volts between terminals or earth
- ✓ Working on a project that USES voltages below 30 Volts (whether generated or transformed).
- ✓ Working on a project that produces/uses Radio frequency signals below 1mW/cm^2
- ✓ Using a P.C. solely to write software/documentation
- ✓ Using lasers classified as Class 1
- ✓ Only Soldering in a designated area, with fume extraction & using colophony free solder
- ✓ Only using small hand-tools to construct electronic circuits (Pliers, wire cutters, and screwdrivers)
- ✓ Working with items at temperatures between 5 degrees and 30 degrees Celsius
- ✗ Not working with revolving or moving parts
- ✗ Not using Chemicals (Including proprietary substances such as glue, cleaning solvents or water)
- ✗ No Eating, Drinking and open toe sandals

Any work outside these parameters must not be carried out until a risk assessment has been made.

In some cases, the risk assessment may be simple clarification from your supervisor or the person responsible for the working area, for example, does a piece of battery operated equipment require testing for electrical safety.

In ALL other cases a full risk assessment will need to be carried out detailing the Hazard and likelihood of the risk, this may need to take into account legislation such as the Electricity at work regulation, The COSHH (Control of substances Hazardous to Health Regulations etc) and the School / University Health and Safety policies.

By signing this form, you are

- a) Stating that all of your work confirms to the above parameters
- b) Stating that you have read the relevant School and University Health and Safety Policies.

This form must be signed, or a written Risk assessment carried out and presented to your supervisor before you start any work and a new risk assessment made whenever circumstances change

Access into swipe operated laboratories will not be allowed until a written risk assessment is produced.

The Student's Project Laboratory 125 is generally designed for the use of UG and MSc who have signed this form and do not require a written risk assessment. However, some activities outside these parameters may be allowed following a risk assessment.

UG & MSc students operating outside these parameters must work in a designated research area and must display a copy of their risk assessment prominently at their working position.

Appendices

I agree that I have assessed my work, and it only involves those operation stated above
I understand that should my work change such that I am operating outside the above
parameters I will need to re-asses the risks.

Location: ____Y8 Engineering Building____ **Supervisor Name:**
____Tim Jackson____

Student Name: ____Joel Ayuba____ **Supervisor Signature:**
____Tim Jackson____

Student ID Card: ____2644630____ **Date:** ____16th
October, 2025____

Student Signature: ________

Date: ____16th October, 2025____

Appendices

Department of Electronic, Electrical and Systems Engineering UNIVERSITY OF BIRMINGHAM GENERAL ETHICAL QUESTIONNAIRE FOR ALL STUDENTS	
Name of student	Joel Ayuba
Email address of student	jia330@student.bham.ac.uk
Name of supervisor	Tim Jackson
Title of Research Project:	Design and Build of a Dual-Core Music Synthesizer for Educational Enhancement
Will the research project involve humans as participants of the research (with or without their knowledge or consent at the time)? This will include any survey, interview or questionnaire that human participants may be asked to complete at any stage as the main part of the project or in the evaluation of the results/deliverables of the project. It will also include any testing of devices, software and other deliverables, which may arise as a result of a project and involves human participants other than the student undertaking the project. Analysis of images or other recordings of human participants or their property and personal possessions is also included.	No
Are the results of the research project likely to expose any person to physical or psychological harm? (Note, before starting the project you will need to complete a risk assessment in all cases)	No
Will you have access to personal information that allows you to identify individuals, or to corporate or company confidential information (that is not covered by confidentiality terms within an agreement or by a separate confidentiality agreement)?	No
Does the research project present a significant risk to the environment or society?	No
Are there any ethical issues raised by this research project that in the opinion of your supervisor require further ethical review?	No

You have answered NO to all of the above questions. Further ethical review is not necessary. You should have this form available at the bench inspections and include it in your final report.

Appendix H: Generative AI Appendix

Contributions of Generative AI to this submission.

There are ethical ways of using Generative AI, examples of which are given at the following link:

[Student and PGR guidance on using GenAI tools ethically for work](#)

Have you used Generative AI in any of the following ways as part of the completion of this assignment (delete as applicable)?

- Scoping out a topic or develop my initial ideas. **(No)**
- Summarising and understand research **(No)**
- To generate code which I have used in my assignment or project **(No)**
- Editing and proofreading my work **(No)**
- I have used it in other ways, not included above to support my work **(No)**